Session 1: Recent Themes in Macroeconometrics

Modern Macroeconometrics

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Why do we need modern macroeconometric methods?

Motivation

- Traditional time series econometric tools were developed for an environment with **just a few series and stable economic relationships**
- Two developments mean that these tools are **no longer sufficient on their own**:
  - **Big Data**: Common to specify models with dozens or even hundreds of variables, as large and rapidly growing number of macroeconomic and financial time series available to researchers and policymakers
  - **Nonlinearities**: There is strong empirical evidence that many macroeconomic time series exhibit **nonlinearities** (e.g., global financial crisis (GFC), zero lower bound (ZLB), Covid pandemic)
- Modelling techniques for handling **heteroskedasticity, outliers, and risk** are crucial in any applied macroeconometric toolkit (in particular, post-Covid)

Motivation (cont.)

- Different time series and transmission channels may feature **different types of nonlinearities** each requiring **different assumptions** about the **underlying functional form**
- Appropriate machine learning (ML) models can uncover **highly complex** relationships arising from a rapidly changing economic environment and structural change
- **Modern Bayesian ML approaches (let the data decide)**
  - **Parametric models** with shrinkage/regularisation and possible time variation
  - **Nonparametric models** that allow for nonlinear relations without imposing a specific functional form

A few things to bear in mind ...

- **Key challenge for ML in macro**: ML methods are typically designed for **large cross-sectional datasets**, but macro data are **"fat" and serially correlated, not "tall" and independent** (many variables, few time periods)
- **Beyond point estimates**: Good policy requires not only accurate predictions but also reliable **uncertainty and (tail) risk assessment** surrounding those
- **Model uncertainty in rich data environments**: With many predictors, any single (linear) model is unlikely to be flexible or robust enough

Big Data in macro

How to best use Big Data information in econometric models?

Big Data and handling many predictors

- Consider a linear regression (K predictors and T observations):
  $y_t = \boldsymbol{x}_t' \boldsymbol{\beta} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma)$
- **"Fat" data**: Many predictors relative to observations (often $K \gg T$)
  - **Example:** FRED-QD with $N = 248$ quarterly US macroeconomic indicators and $T \approx 240$ time periods
  - Common practice to use P lags of the predictors, i.e., $K = N \cdot P$
- Even with $K < T$, estimation is prone to **overfitting**: The model fits noise rather than signal; **poor out-of-sample predictions** as a result
- **Regularisation** imposes structure to separate signal from noise by penalising large coefficients, **shrinking irrelevant predictors** towards zero (Session 2)

So which predictors carry signal then?

The model space is vast!

- With K potential predictors, there are 2^K possible regression models (each variable is either included or excluded)
  - Say for $K = 100$ we have 2^{100} possible models
- Two Bayesian (data-driven) approaches commonly applied in the empirical macro literature:
  - **Bayesian Model Averaging (BMA)**: Estimates inclusion probabilities across all K predictors; considered a gold standard but computationally demanding and assumes a single "true" model exists within the model space
  - Regularisation done quite naturally in the Bayesian framework via so-called "prior"
  - **Shrinkage priors**: Regularisation via priors on β that approximate discrete variable selection continuously; more ML-flavoured and scalable, but no inclusion probabilities

Nonlinearities in macro data

Nonlinearities can take many forms; how to effectively capture them?

Vast evidence for instability in macro relationships

- Macro relationships are **not stable** over time; both **constant parameters** β and **homoskedastic errors**, $\varepsilon_t \sim \mathcal{N}(0, \sigma)$, are often unrealistic
  - **Phillips curve**: Flattened substantially in the 2010s, re-emerging in the 2020s
  - **Monetary transmission**: Varies across the business cycle and across policy regimes
  - **Macro volatility / uncertainty**: Low and stable during the Great Moderation, elevated during GFC, and sharply spiked during Covid
  - **Tail risks**: Asymmetric downside/upside risks to growth and inflation are time-varying and state dependent
- These facts call for more flexible modelling of:
  - **Transmission channels**: Via time-varying coefficients β_t or nonlinear functions $f(\boldsymbol{x}_t)$
  - **Shock volatilities**: Via heteroskedastic errors $\varepsilon_t \sim \mathcal{N}(0, \sigma_t)$

Modelling nonlinearities

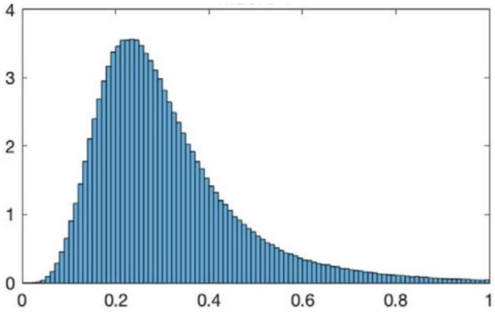
- The approaches we will consider to capture potential nonlinearities
  - **Modern TVP regressions (Session 3)**: We let the data decide which coefficients actually move; whether coefficients vary over time is treated as a model selection problem (static versus dynamic sparsity)
  - **Heteroskedasticity (Session 4)**: Modelling volatility clustering, inflated predictive uncertainty during turbulent times, absorbing exceptional outliers (e.g., Covid), and avoiding bias in transmission channels (Sims critique)
  - **Quantile regressions (Session 5)**: Capturing the macroeconomy at risk, monitoring downside risks (growth) or upside risks (inflation)
  - **Nonparametric ML methods (Session 6)**: Bayesian additive regression trees, Gaussian processes, and neural networks are agnostic about functional form and let the data decide on the functional form

Few motivating examples from the empirical macro literature

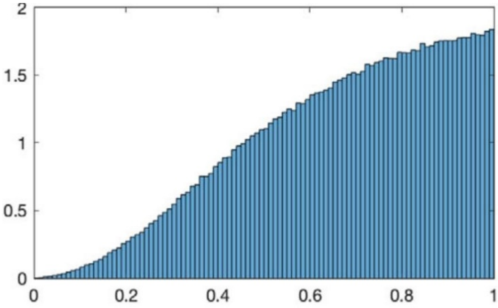
Big Data in macro: Illusion of sparsity

- When many predictors are available, should we assume the "true" model is **sparse** (few relevant predictors) or **dense** (many small effects)?
- Giannone, Lenza & Primiceri (2021, *ECTA*)
  - Specify a prior allowing for both variable selection and shrinkage
  - Dense representations typically favoured, suggesting sparsity in macro data may be an **illusion**
- Next two slides show applications to macro and financial data from their paper

Estimated degree of sparsity in two macro applications



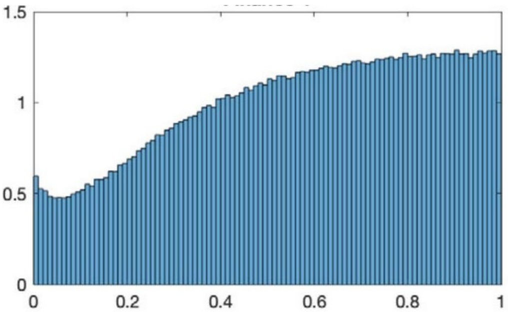
Macro 1: US industrial production forecasting (K = 130 predictors); intermediate between sparse and dense



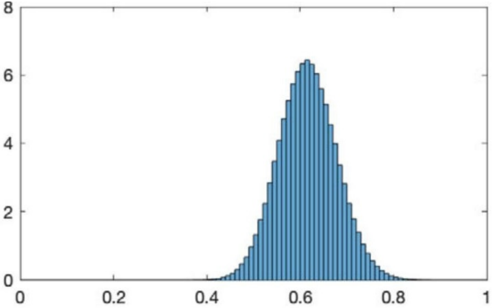
Macro 2: Cross-country GDP growth determinants (K = 60 predictors); strongly favours dense representations

Source: Giannone, Lenza & Primiceri (2021, *Econometrica*), Figure 4

Estimated degree of sparsity in two finance applications



Finance 1: US equity premium prediction ($K = 16$ predictors); little information about model size



Finance 2: Cross-section of US stock returns ($K = 144$ predictors); intermediate/dense, not sparse

Source: Giannone, Lenza & Primiceri (2021, *Econometrica*), Figure 4

Big Data in macro: Illusion of sparsity (cont.)

- Fava & Lopes (2020, *arXiv*)
  - The illusion of sparsity may itself be an illusion
  - Results are sensitive to the prior on regression coefficients (prior sensitivity analysis)
- Gruber & Kastner (2025, *IJoF*)
  - Sparse vs dense priors in Bayesian vector autoregressions (VARs)
  - Forecasting performance varies across variables and over time $\longrightarrow$ **it depends!**

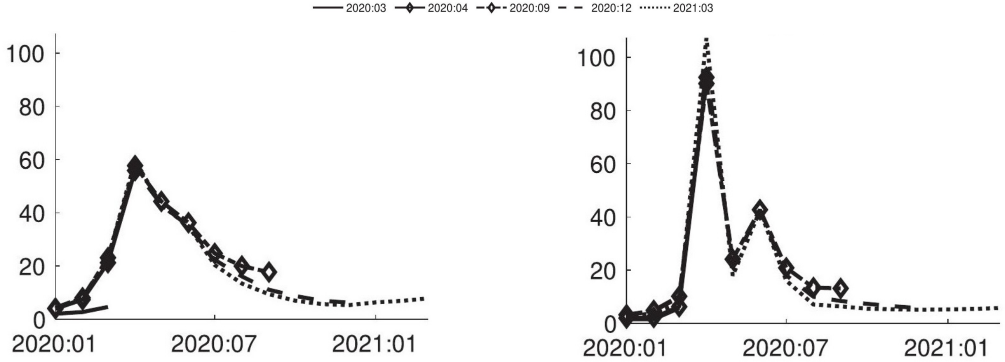
The pandemic: A stress test for macro models

- The pandemic provided an extreme test for our macroeconometric models
- Standard linear models with **constant parameters** and **homoskedastic errors** cannot cope with such extreme observations
- Was Covid an outlier or a structural break? Evidence suggests treating it as an **elevated volatility episode** is sensible
  - Real activity and labour market variables massively affected in 2020
  - Post-pandemic fluctuations (e.g., surge in inflation) are features to **model**, definitely not outliers to discard

Heteroskedasticity and outlier modelling in macro

- **Stochastic volatility (SV)**
  - Time-varying error variances absorb extreme observations
  - Treats GFC and Covid as elevated volatility rather than structural breaks
  - Gold standard in empirical macro (see Chan (2024) for an overview)
- **Lenza & Primiceri (2022, *JAE*)**
  - How to estimate a vector autoregression (VAR) after March 2020?
  - More or less amounts to dropping Covid observations
- **Cascaldi-Garcia (2025, *WP*)**: Pandemic priors for Bayesian VARs
- **Carriero, Clark, Marcellino & Mertens (2024, *ReStat*)**: SV-augmented Bayesian VAR that is able to handle Covid outliers without ad hoc data adjustments

Covid as a volatility episode: SV vs outlier-augmented SV



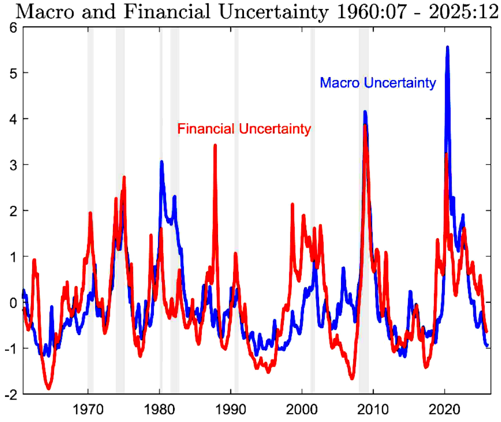
SV: Volatility spikes in April 2020 and remains elevated; treats Covid as a **more persistent** shift

SVO-t: Volatility spikes sharply but quickly reverts; identifies Covid as a **transitory** outlier

Source: Carriero, Clark, Marcellino & Mertens (2024, *ReStat*), Figure 2 (payroll growth volatility)

Measuring uncertainty

Jurado, Ludvigson & Ng (2015, *AER*)



Source: Sydney Ludvigson's macro and financial uncertainty indices

Macro and financial uncertainty indices spike sharply during recessions and/or financial crises; when SV models assign higher σ_t^2

Beyond point forecasts

Why density forecasts?

- Good forecasting analysis is not only about getting the point estimate right
  - A point forecast for inflation or GDP growth has little practical value if the range of possible outcomes is extremely wide, e.g., anywhere between $+/-10\%$
  - Closely related to this concern is the growing focus on tail risks
- It is equally important to consider the uncertainty surrounding those estimates
- A h-step ahead **density forecast** of y_{T+h} conditional on the information set $\mathcal{I}_{1:T}$ provides the full predictive distribution:

$$p(y_{T+h}|\mathcal{I}_{1:T})$$

  - Quantifies uncertainty around a point forecast; key for risk management
  - Allows computation of probabilities, say the probability of negative GDP growth $\Pr(y_{T+h} < 0|\mathcal{I}_{1:T})$

Why stochastic volatility and outlier adjustments matter?

- **More credible density forecasts**: Uncertainty bands that realistically widen in turbulent periods and narrow in tranquil ones
- Captures a simple but important notion: When macroeconomic uncertainty is high, this should be reflected in our confidence about future outcomes
- An increase in σ_t^2 directly implies an increase in the intrinsic, unforecastable forecast error ϵ_t
- SV models are widely used to construct **uncertainty indices** in macro and finance

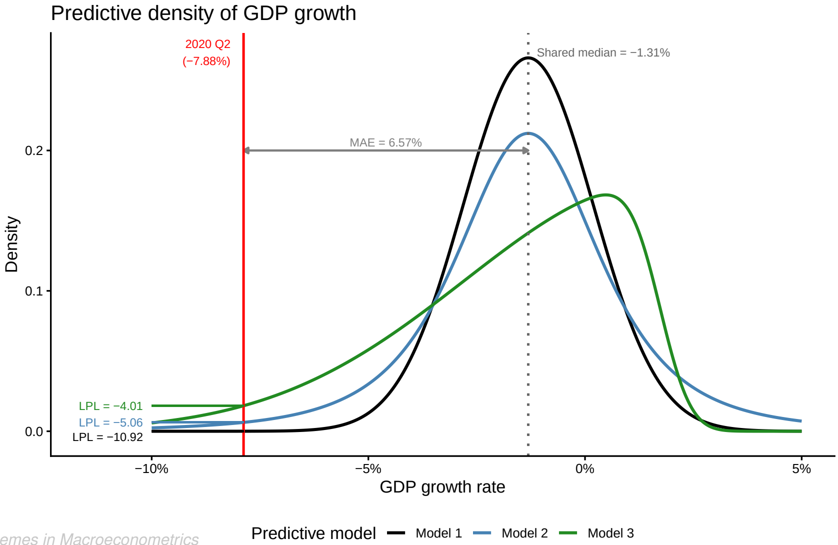
Evaluating density forecasts

- Point forecasts are evaluated with root mean square forecast errors (RMSFEs) or mean absolute errors (MAEs)
- **Log Predictive Likelihood (LPL)**: Evaluates the predictive density at the realised value

$$\mathrm{LPL}(h) = \frac{1}{T_1 - T_0} \sum_{\tau=T_0}^{T_1} \log p\left(y_{\tau+h}^{(r)} \mid \mathcal{I}_\tau\right)$$

  - Higher is better
  - Heavily penalises models that assign low probability to what actually happened

Evaluating density forecasts (cont.)



Evaluating density forecasts (cont.)

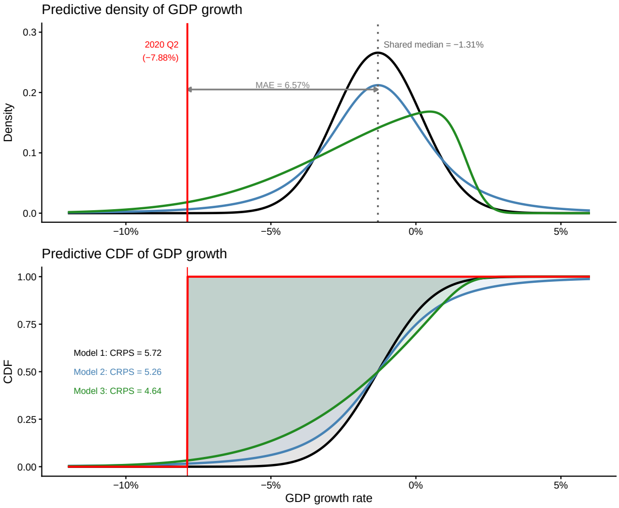
- **Continuous Ranked Probability Score (CRPS)**: Evaluates the full predictive CDF

$$\text{CRPS}(h) = \frac{1}{T_1 - T_0} \sum_{\tau=T_0}^{T_1} \int_{-\infty}^{\infty} \left[F_{\tau+h}(z) - \mathbf{1}\left(y_{\tau+h}^{(r)} \leq z \right) \right]^2 dz$$

  - Lower is better
  - Proper scoring rule, which generalises MAE to densities

- **Tail-weighted CRPS**: Up-weights specific regions of the distribution (important for tail risk applications)

Evaluating density forecasts (cont.)

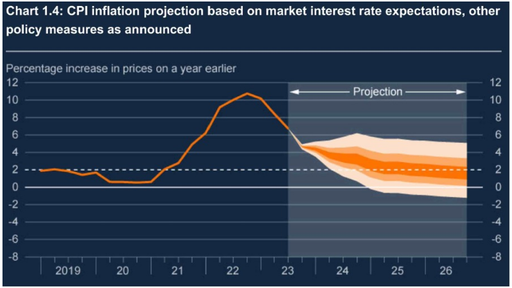


Fan charts and the Bernanke review

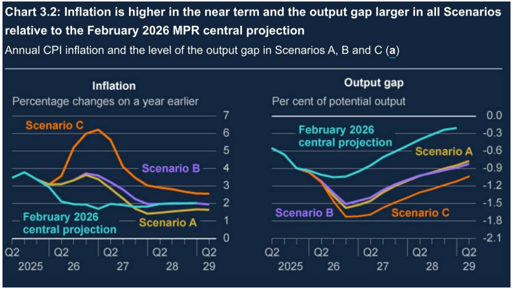
- **Fan charts**: For a long time, central banks' standard tool for communicating predictive distributions; shaded bands show credible intervals of predictive density
- **Bernanke (2024) review** flagged "serious deficiencies" in the Bank of England (BoE)'s forecasting **infrastructure** and also recommended **retiring the fan chart** in favour of scenario analysis
- **Direct consequence**: BoE no longer uses fan charts but **scenario-based projections**, which is also easier to digest for the public audience

BoE before and after Bernanke report

Fan charts ⟶ scenarios



November 2023 MPR: Symmetric fan chart



April 2026 MPR: Three explicit scenarios

Source: Bank of England Monetary Policy Reports, November 2023 (Chart 1.4) and April 2026 (Chart 3.2)

Quantile regression (Session 5)

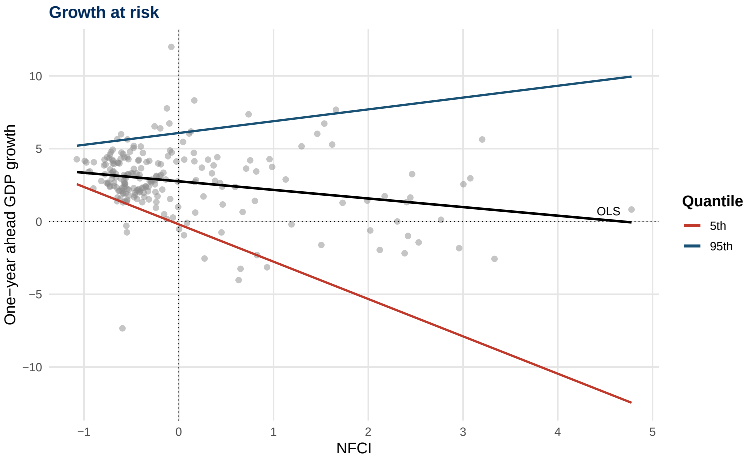
- **OLS** estimates the conditional **mean**: $\mathbb{E}[y_t|\boldsymbol{x}_t] = \boldsymbol{x}_t'\hat{\boldsymbol{\beta}}$ by minimising squared residuals (one line through the data)
- **Quantile regression (QR, Koenker & Bassett, 1978, *ECTA*)** targets the conditional **quantile** $Q_\tau(y_t|\boldsymbol{x}_t) = \boldsymbol{x}_t'\hat{\boldsymbol{\beta}}_\tau$, with a τth quantile-specific coefficient
- Running QRs at many values of $\tau \in (0, 1)$ allows us to recover a **full conditional distribution**
- Applications in macroeconometrics: **Growth at risk**, **inflation at risk**, and many more (see next slides)

Macro at risk

Growth at Risk (GaR)

- **Main takeaway** of Adrian, Boyarchenko & Giannone (2019, *AER*):
  - Financial conditions shift the lower quantiles ($\tau = 0.05, 0.10$) of GDP growth sharply but barely move the median
  - Heterogeneity in $\boldsymbol{\beta}_\tau$ across quantiles; tail risk and heteroskedasticity as a function of covariates
  - OLS misses this: A single line cannot capture the **asymmetric response** of the distribution

Growth at Risk (GaR)

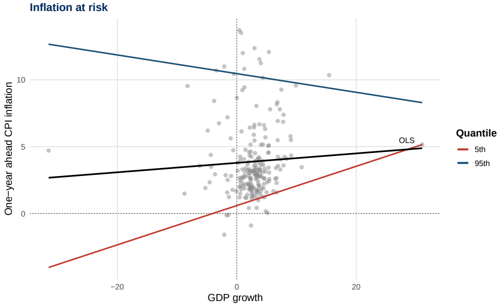


Inflation at Risk (IaR)

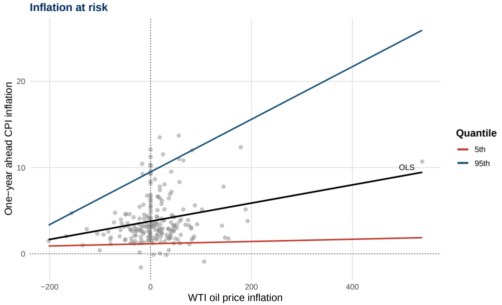
- **Inflation at risk**: Apply QR to one-year-ahead CPI inflation on WTI oil price inflation and GDP growth
- **Downside risk** (left): Deflation risk in case of demand shocks
- **Upside risk** (right): Inflation risk in case of supply shocks; surging oil prices dramatically raise the right tail of the inflation distribution
- Again, OLS captures only the average effect and masks the **asymmetry** across the distribution

Inflation at Risk (IaR)

(a) Downside risk



(b) Upside risk



ML in macroeconometrics

Frontier research and possible way forward

Outlook Session 6: ML meets macroeconometrics

- Parametric nonlinear models make specific assumptions about the **form of the nonlinearity**, which may itself be misspecified
- **Machine learning** methods relax this; they let the data determine the functional form $f(x_t)$ with minimal assumptions
- These approaches borrow from the statistical learning literature and are increasingly paired with classical time series models in macroeconometrics
  - **Bayesian additive regression trees (BART)**: a sum of regression trees, each capturing a different nonlinear fragment; Bayesian regularisation prevents overfitting (Chipman, George, and McCulloch, 2010, *AOAS*)
  - **Gaussian processes (GP)**: flexible priors over functions; agnostic on the functional form between predictors and the target (Rasmussen and Williams, 2006)
  - **Bayesian neural networks (BNN)**: can approximate any nonlinear function arbitrarily well (Hornik et al., 1989)

The way forward: GenAI meets macroeconometrics

- A newer wave of ideas from the GenAI literature is starting to reshape macroeconometric modelling and forecasting
  - **Benign overfitting**: ML models show good out-of-sample predictions even with a perfect fit
  - **Attention-based vector autoregressions**: Attention (the engine behind LLMs) produces state-dependent, time-varying dynamics that remain deterministic functions of past data
  - **Time series foundation models**: Pretrained on millions of diverse time series, they forecast zero-shot on a new dataset, with no task-specific fitting

Being a Bayesian macroeconometrician

Why Bayesian?

- Lots of work in macroeconomics and forecasting uses Bayesian methods nowadays
- **Bayesian econometrics** is naturally well suited for macro applications
  - **Regularisation**: Shrinkage information via the prior (essential with small T)
  - **Principled and coherent uncertainty quantification**: The posterior characterises full parameter uncertainty, which propagates into the predictive distribution; posterior predictive distributions arise naturally
  - **Flexible, modular modelling**: A Bayesian model is like a box of tools that can be assembled according to our needs; e.g., we might start with a simple regression model and then add a hierarchical prior structure for shrinkage, or augment an SV model with an outlier adjustment via data augmentation

Statistical software used by Bayesians

- Most Bayesians write their own codes (e.g., using Matlab, Julia, Python, R or C++) to do posterior simulation
- Bayesian work cannot (easily) be done in standard econometric packages like Stata (Session 2 explains why)
- Stata has some Bayes, but limited (and still little for macroeconomics)
- Researchers typically post their code on their webpages or GitHub
  - See my GitHub page here (R)
  - Colleagues and co-authors who provide excellent code packages: Joshua Chan (Matlab), Florian Huber (R), Karin Klieber (R), Gary Koop (Matlab), Dimitris Korobilis (Matlab), Haroon Mumtaz (Matlab), Michael Pfarrhofer (R), Aubrey Poon (Matlab), Tomasz Woźniak (R)